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Manual for HuBase™:

HuBase™ Functionality Summary:

1. Pharmacology: HuBase™ allows you to examine the pharmacology information of PDX models.
2. Gene Expression: HuBase™ allows you to compare the expression level of a single gene (or simultaneously compare upto eight genes) on a selection of models.
3. Gene Copy Number: HuBase™ allows you to examine the gene copy number of a single gene over a selected number of models.
4. Gene Mutation: HuBase™ allows you to examine the mutation status of selected genes.
5. MiRNA gene expression: HuBase™ allows you to examine the miRNA gene expression on selected models.
6. Gene fusion: HuBase™ allows you to search gene fusion in different models or all fusion genes in one selected model.
7. HuBase™ allows you to search PDX models according to your search conditions.

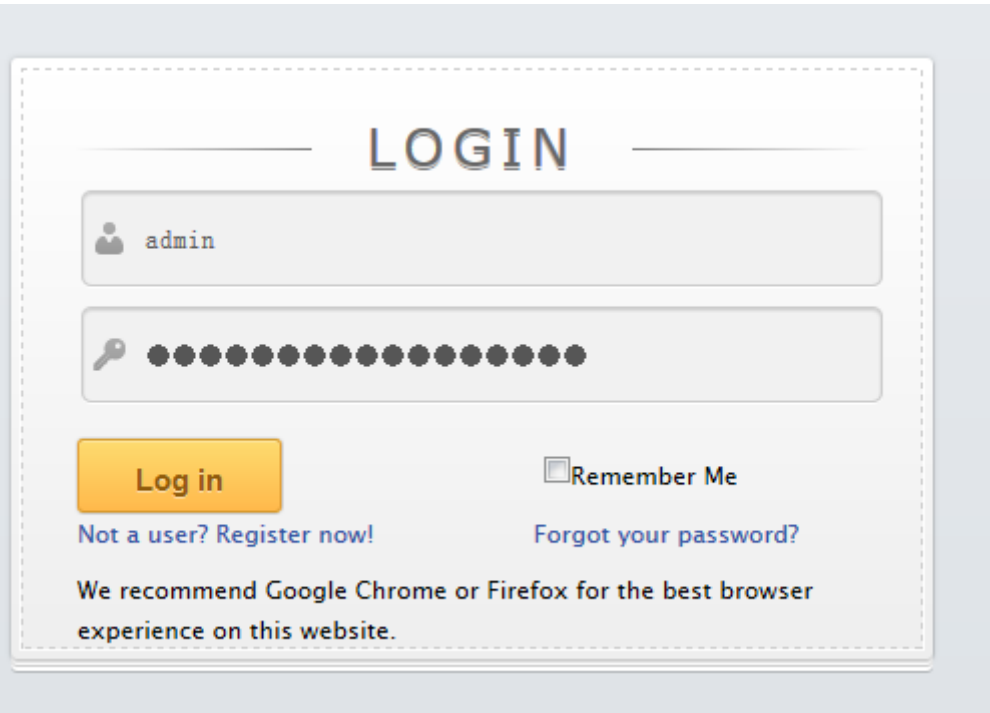
This manual is written for the first time users of HuBase™, and tells users how to perform the queries step by step.

The content of this manual is organized by type of queries. One query type corresponds to one subtitle.

To ensure maximal functionality is displayed, please use google chrome or firefox and update to the latest version of Adobe Reader when logging into HuBase™.

1.Log in

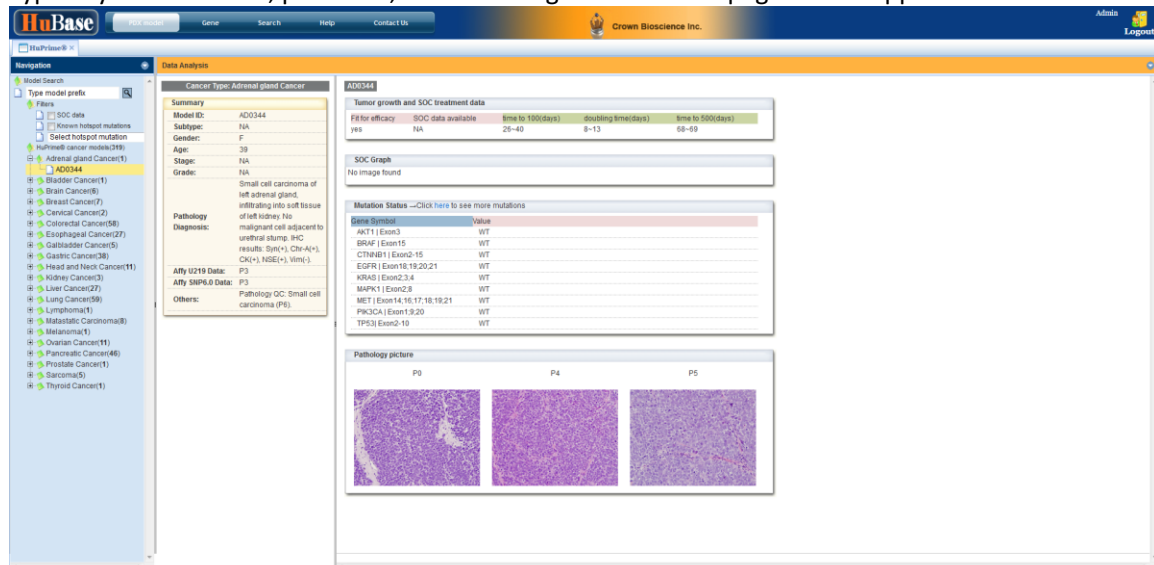
Type "http://HuBase.crownbio.com " into the address bar of the browser, the login page should appear as below:



The login page features a central dashed box with the word "LOGIN" at the top. Below it are two input fields: the first contains the username "admin" and is preceded by a person icon; the second is a password field with a key icon and a series of dots. To the right of the password field is a "Remember Me" checkbox. Below these fields is an orange "Log in" button. Further down are two links: "Not a user? Register now!" and "Forgot your password?". At the bottom, a recommendation states: "We recommend Google Chrome or Firefox for the best browser experience on this website."

If "Remember Me" checked, you will stay logged in for 1 week.
You can click "Forgot your password" to retrieve it if you forget your password.

Type in your username, password, and click "Log in". the home page should appear as below:



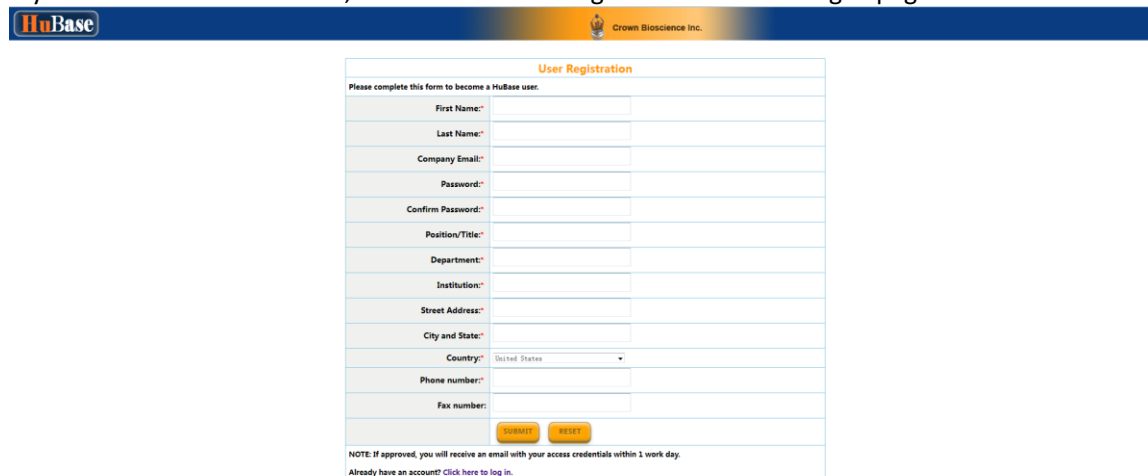
The home page has a blue header with the "HuBase" logo and navigation links. A left sidebar lists various cancer types. The main content area is titled "Data Analysis" and shows details for "Cancer Type: Adrenal gland Cancer" and "Model ID: ADO344". It includes a "Summary" section with patient information, a "Pathology" section with a description, and a "Mutation Status" table. At the bottom, there are "Pathology picture" thumbnails labeled P0, P4, and P5.

Gene Symbol	Value
AKT1 (Exon3)	WT
BRCA1 (Exon15)	WT
CTNNB1 (Exon2-15)	WT
EGFR (Exon18,19,20,21)	WT
KRAS (Exon2,3,4)	WT
MAPK1 (Exon2,8)	WT
MET (Exon14,15,17,18,19,21)	WT
PRKCA (Exon1,9,20)	WT
TP53 (Exon2-10)	WT

You can change your personal information by clicking the top right corner of your name, the personal information includes your password , position, department, street address, phone number, and so on.

2.Register

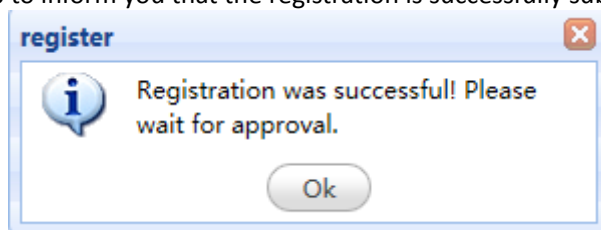
If you do not have an account, click "Not a user? Register now!" in the log in page to create one:



The image shows the HuBase User Registration form. At the top, there is a blue header with the HuBase logo on the left and the text "Crown Bioscience Inc." on the right. Below the header, the form is titled "User Registration" in orange. The instruction "Please complete this form to become a HuBase user:" is displayed. The form contains the following fields: First Name, Last Name, Company Email, Password, Confirm Password, Position/Title, Department, Institution, Street Address, City and State, Country (a dropdown menu currently showing "United States"), Phone number, and Fax number. At the bottom of the form, there are two orange buttons labeled "SUBMIT" and "RESET". Below the buttons, a note states: "NOTE: If approved, you will receive an email with your access credentials within 1 work day. Already have an account? Click here to log in."

Type in your information, then click "submit" to submit your information (If approved, you will receive an email with your access credentials within 1 work day.).

A dialog box will pop up to inform you that the registration is successfully submitted:



3.PDXmodel- HuPrime®

Select one PDX model after the prompt of "Model Search", then click  , the Pharmacology information will show at the right side.

Navigation

- Model Search
 - Type model prefix
 - Filters
 - ☒ SOC data
 - ☐ Known hotspot mutations
 - Select hotspot mutation
 - HuPrime® cancer models(33)
 - Breast Cancer(1)
 - Colorectal Cancer(11)
 - Esophageal Cancer(4)
 - Gastric Cancer(11)
 - Head and Neck Cancer(1)
 - Kidney Cancer(1)
 - Liver Cancer(4)

Data Analysis

Cancer Type: Adrenal gland Cancer

Summary

Model ID:	AD0344
Subtype:	NA
Gender:	F
Age:	39
Stage:	NA
Grade:	NA

Pathology:
Diagnosis: Small cell carcinoma of left adrenal gland; infiltrating into soft tissue of left kidney; NA malignant cell adjacent to urethral stump; IHC results: Syn(+), Cdx-A(+), CK(+), Ki67(+), Vimentin(+).

Atty U219 Data: P3
Atty SHP6.0 Data: P3
Others: Pathology QC: Small cell carcinoma (P8)

AD0344

Tumor growth and SOC treatment data

Eff for efficacy	SOC data available	Time to 150(days)	Shrinking time(days)	Time to 250(days)
yes	NA	20-40	8-13	68-69

SOC Graph
No image found

Mutation Status —Click here to see more mutations

Gene Symbol	Value
AKT1 Exon3	WT
BRAF Exon15	WT
CTNNB1 Exon2-15	WT
EGFR Exon18,19,20,21	WT
KRAS Exon2,3,4	WT
MAPK1 Exon2,8	WT
MET Exon14,15,17,18,19,21	WT
PIK3CA Exon19,20	WT
TP53 Exon2-10	WT

Pathology picture

P0 P4 P5

You can constrain the displaying of only models with pharmacologic data by checking the "SOC data" box under Filters prompt.

Model Search

Type model prefix

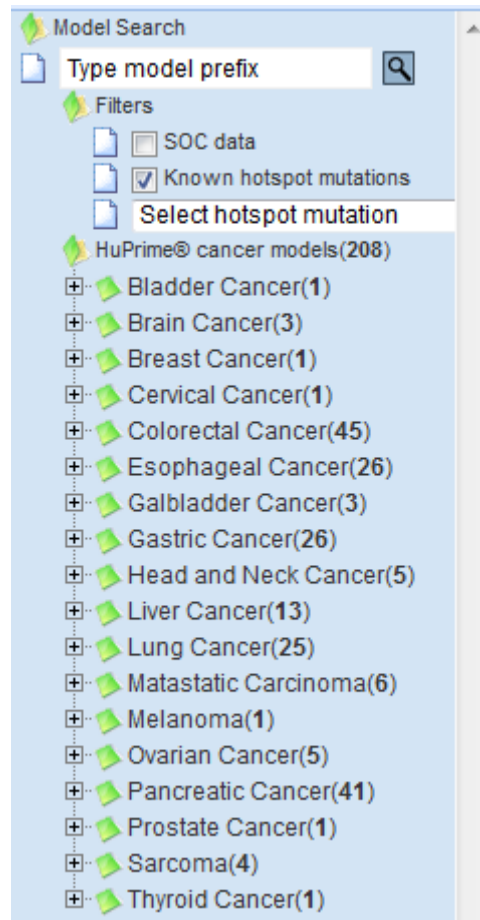
Filters

- ☒ SOC data
- ☐ Known hotspot mutations
- Select hotspot mutation

HuPrime® cancer models(33)

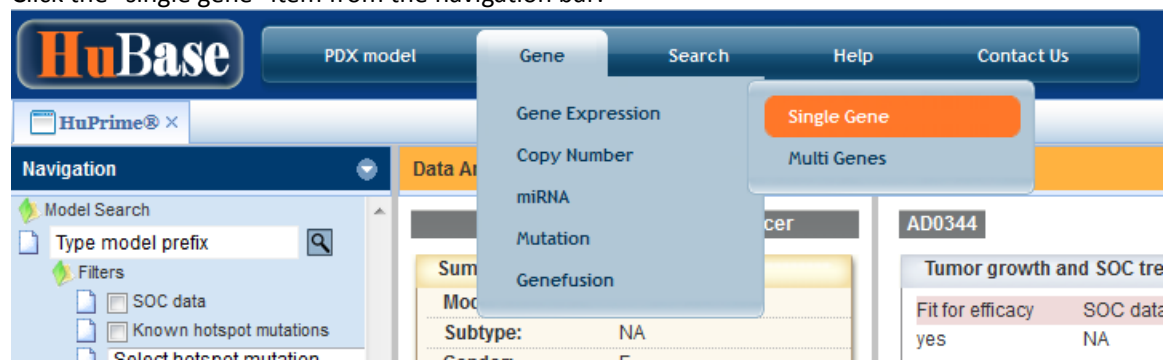
- Breast Cancer(1)
- Colorectal Cancer(11)
- Esophageal Cancer(4)
- Gastric Cancer(11)
- Head and Neck Cancer(1)
- Kidney Cancer(1)
- Liver Cancer(4)

You can display the models only with selected hotspot mutations by checking the "Known hotspot mutations" and selecting a certain mutation type in the following text box.

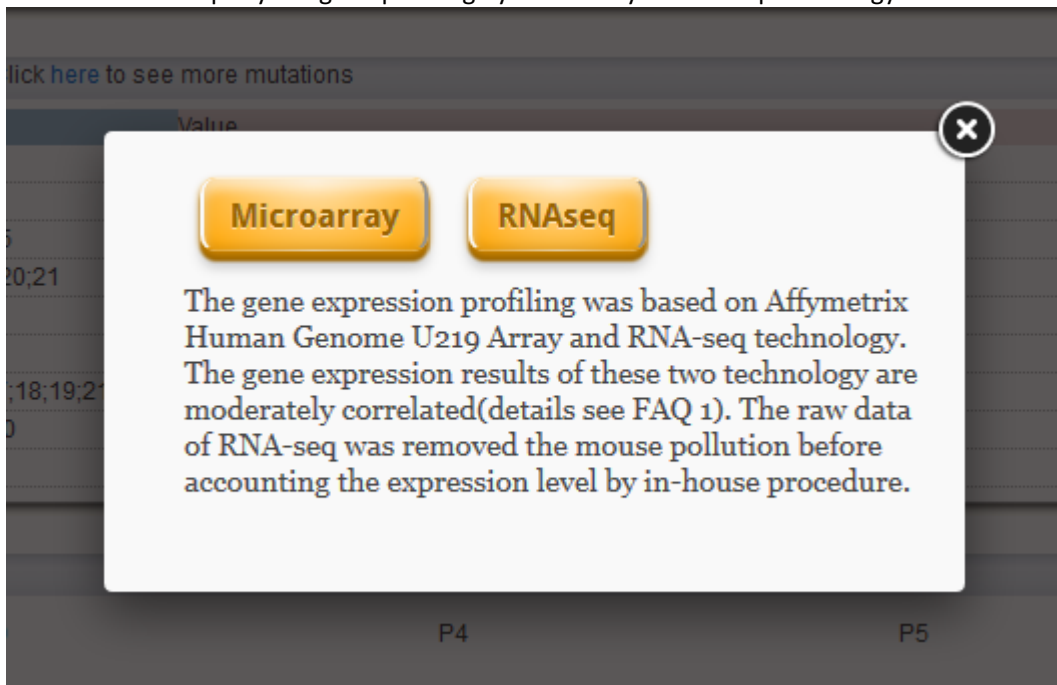


4. Gene-Gene Expression-Single Gene

Click the "single gene" item from the navigation bar:



You can select to query the gene profiling by microarray or RNAseq technology.



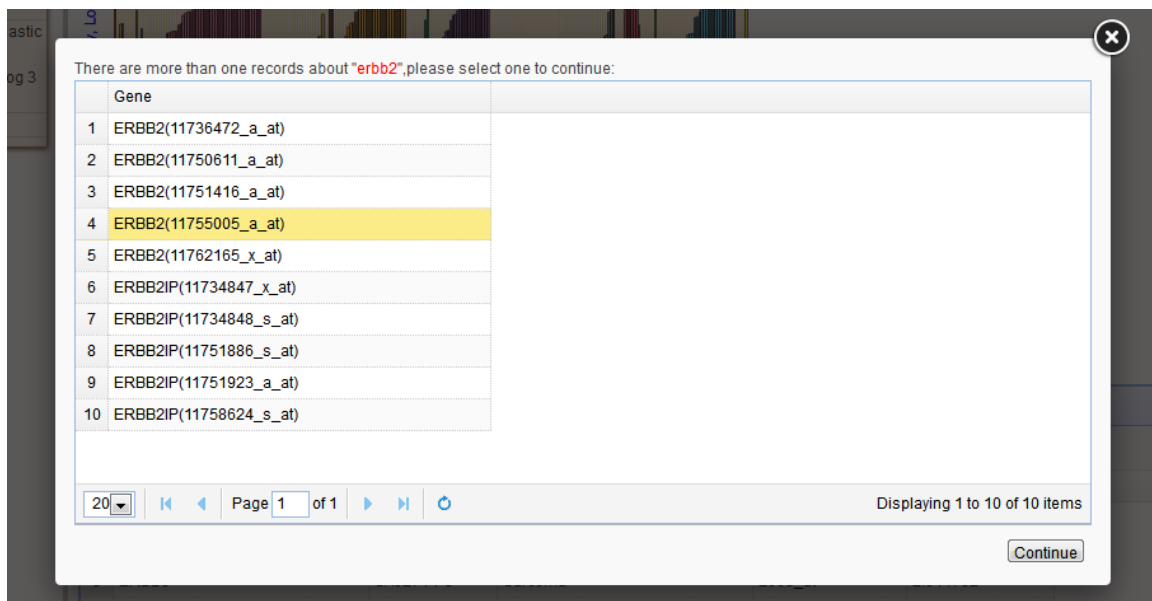
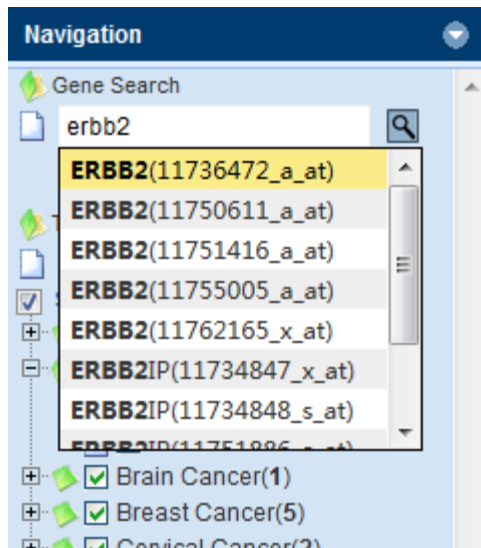
4.1 selecting gene and PDX models:

Type in gene names under the prompt of "Gene:":

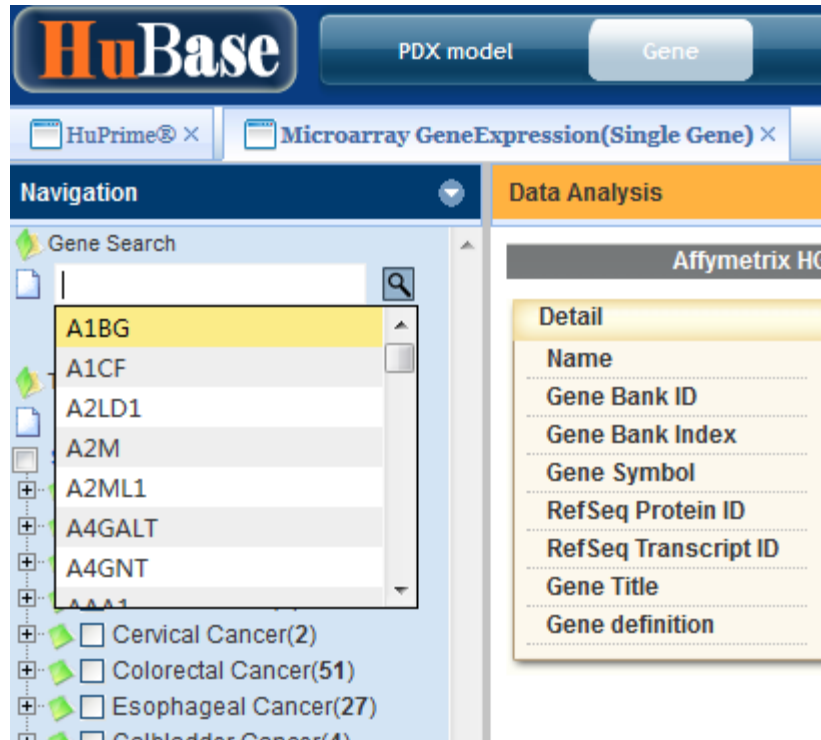
A screenshot of a "Gene Search" form. It has a title "Gene Search" with a green leaf icon. Below the title is a text input field with a magnifying glass icon on the right. At the bottom, there are two radio buttons: "Custom CDF" (which is selected) and "Affymetrix CDF".

For microarray based technology, the gene expression value has been processed using affymetrix CDF and custom CDF, separately. For custom CDF, one gene is corresponding to one probeset, which is easy to handle with. You can select any one dataset as your need.

If you select affymetrix CDF based expression data, you need further select the probeset in the popped page.



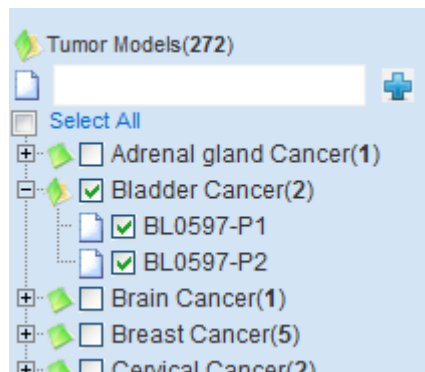
For RNAseq based technology, Both transcript and integrated gene expressions are shown, you can select any type of data as your need.



Select the PDX model category under the prompt of "Tumor models":

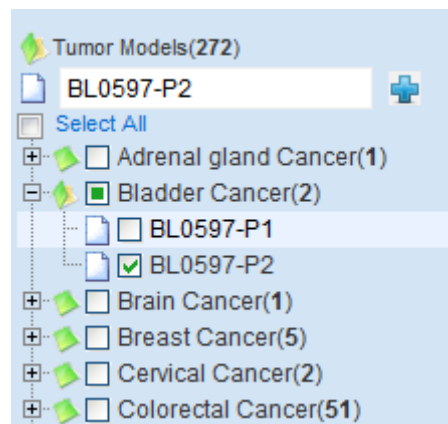
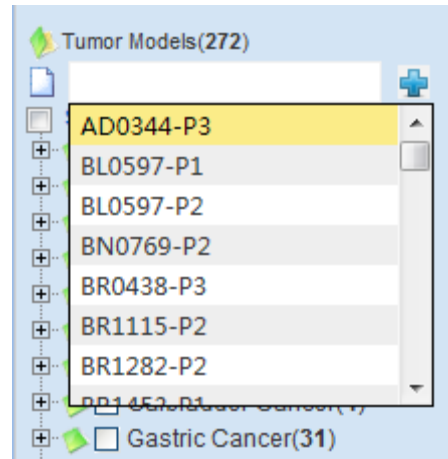
You can select models through the input box or the checkbox:

1. by checkbox:

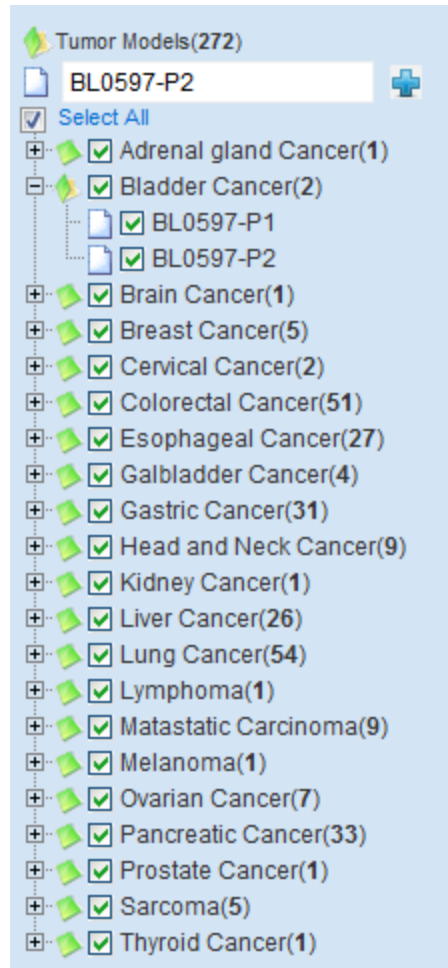


2. by entering the names of models directly: for example, type in "BL0597-P2" in the input box and

click , then the corresponding model will be selected.

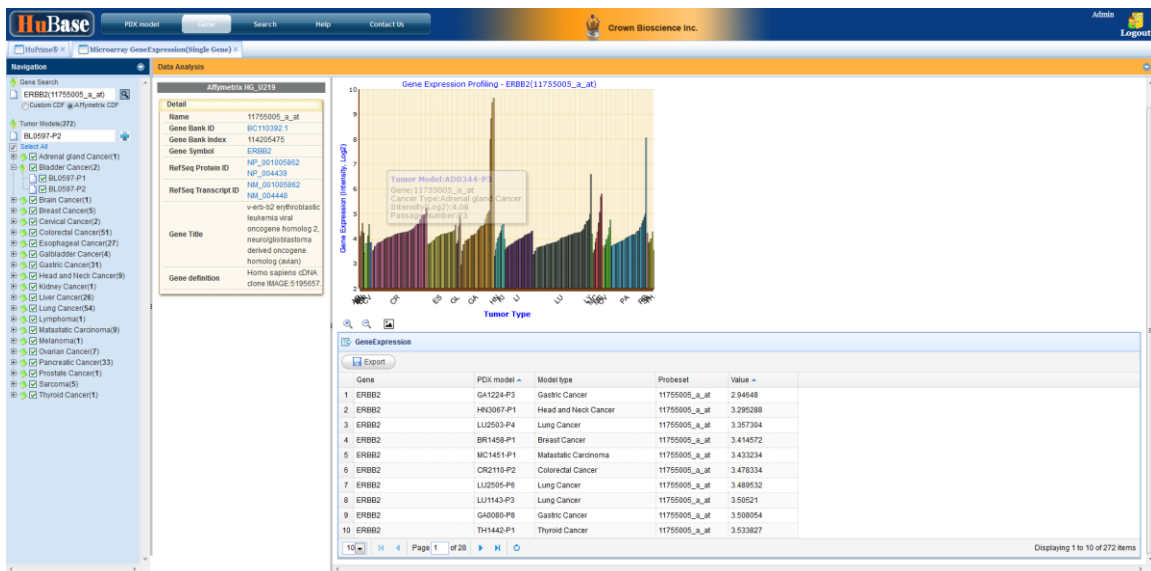


3. select all: you can also tick "select all" to select all the models in the category.

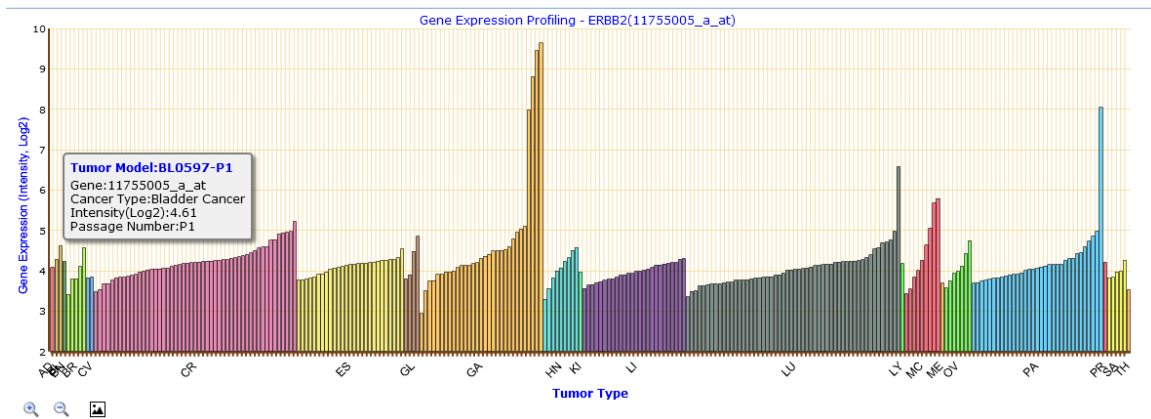


4.2 displaying result

After gene and models are selected, click the button of , then the expression profiling of this gene is displayed:



The chart can be zoomed in or out by clicking the  at the bottom left of the chart, The chart can be exported in the format of png by clicking the button of .



The table can be sorted in ascending or descending order of expression level by click "Value", It can also be sorted in alphabetic order by "PDX model":

PDX model ▲	Value ▲
GA1224-P3	2.94648
HN3067-P1	3.295288
LU2503-P4	3.357304
BR1458-P1	3.414572
MC1451-P1	3.433234
CR2110-P2	3.478334
LU2505-P6	3.489532
LU1143-P3	3.50521
GA0080-P8	3.508054
TH1442-P1	3.533827

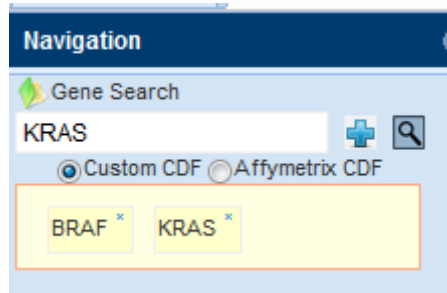
 Export

The table can be exported in xls format by clicking the button of  :

5. Gene-Gene Expression-Multi Genes

Click the "multi genes" item from the navigation bar.

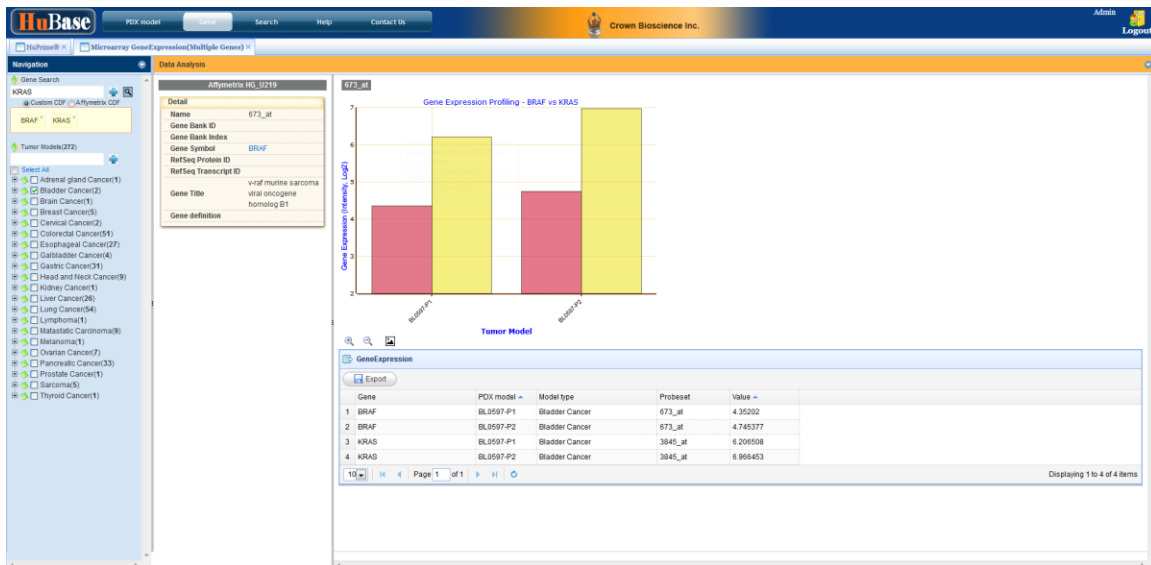
Type in the gene name in the input box under the prompt of "Gene:" and click  to add genes to the select box.



The image shows a 'Navigation' panel with a 'Gene Search' section. A text input field contains 'KRAS'. To the right of the input field are a blue plus icon and a magnifying glass icon. Below the input field are two radio buttons: 'Custom CDF' (selected) and 'Affymetrix CDF'. At the bottom, there is a yellow box containing two gene names: 'BRAF' and 'KRAS', each followed by a small 'x' icon.

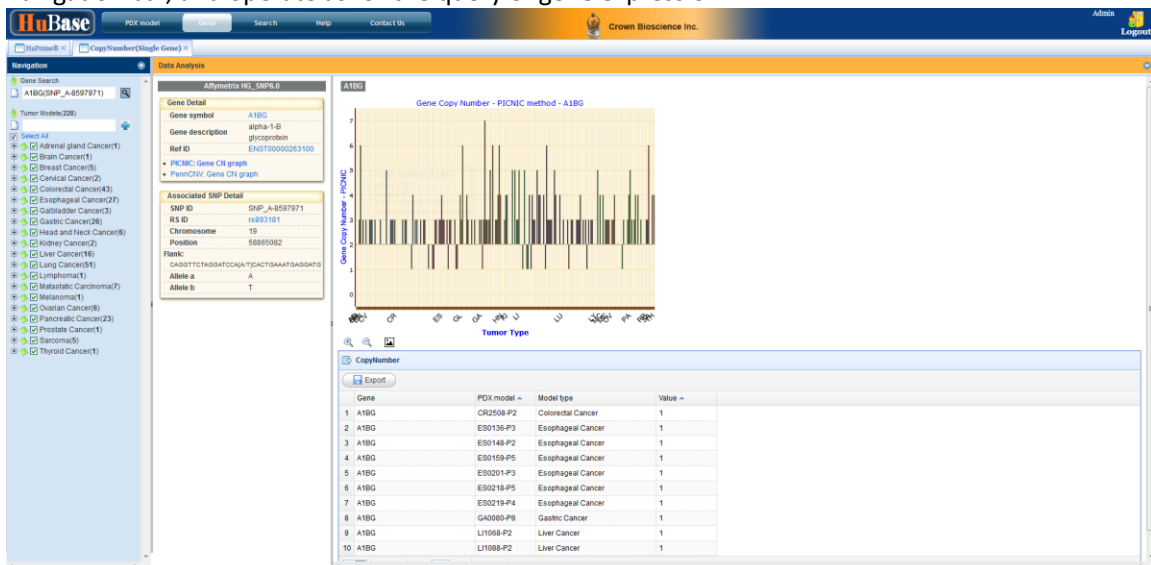
Select models as previously described, click  to display chart and table.

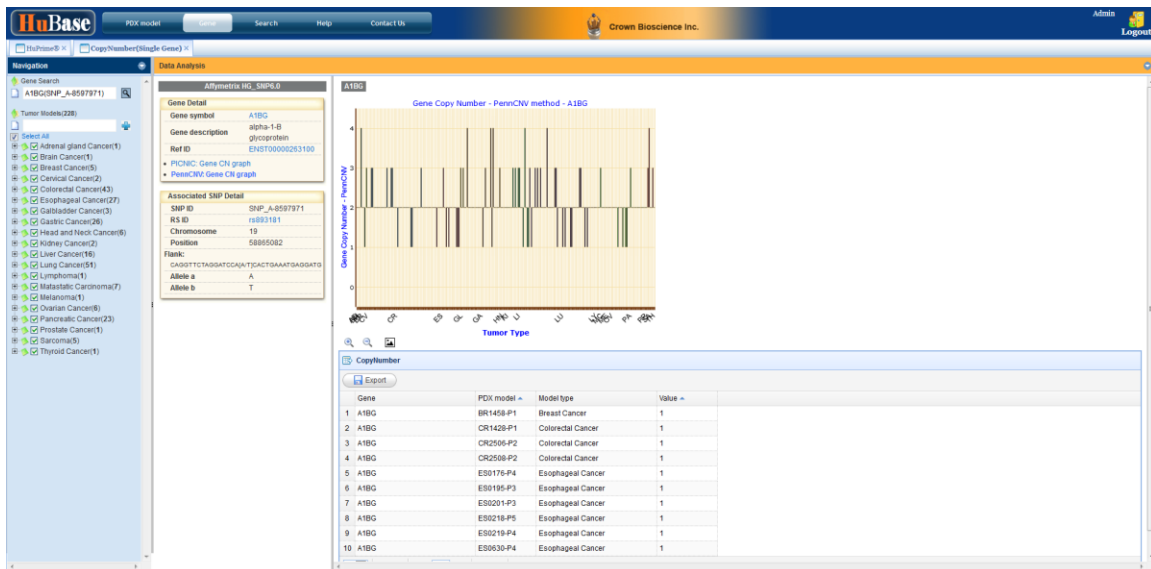
If less than 4 genes were selected, all models can be displayed on the chart, and when no less than 4 genes were selected, at most 100 models should be selected in order to properly display the chart.



6. Gene-Copy Number

Click "Gene->Copy number->Single Gene" or "Gene->Copy number->Multi genes" from the navigation bar, and operate as for the query of gene expression:

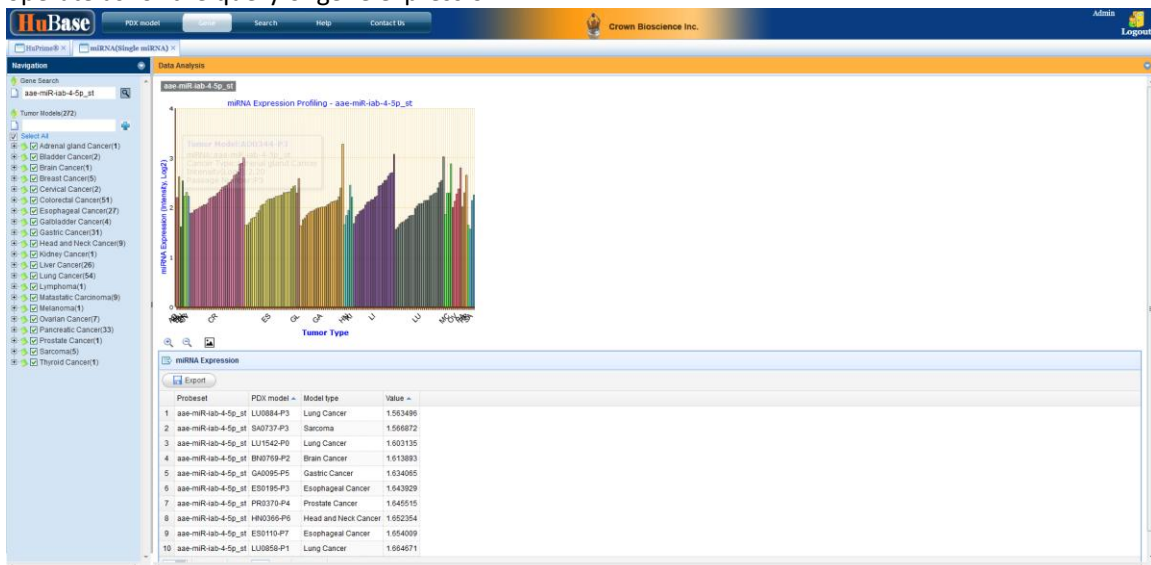




The gene copy number has been detected using PICNIC and pennCNV softwares, separately, you can select the PICNIC or PennCNV as your need by clicking on "PICNIC: Gene CN graph" or "PennCNV: Gene CN graph".

7. Gene-miRNA

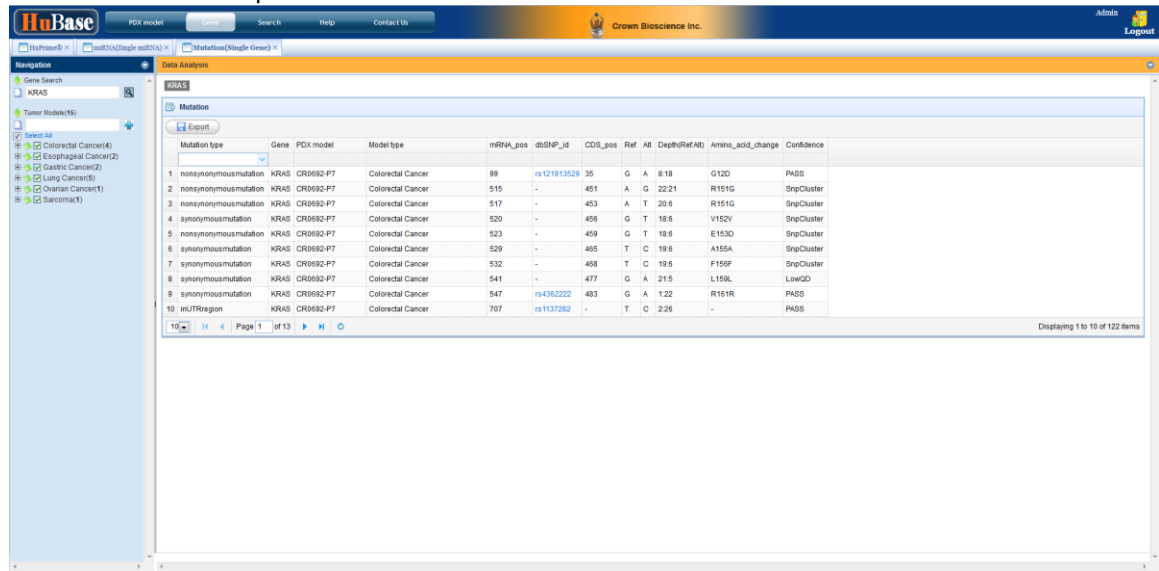
Click "Gene->miRNA->Single Gene" or "Gene->miRNA->Multi genes" from the navigation bar, and operate as for the query of gene expression:



8. Gene-Mutation

Click "Gene->Mutation->Single Gene" or "Gene->Mutation->Multi Genes" from the navigation bar, select a gene/genes and PDX models, click  to display mutation information, and you can click

 **Export** to export result in xls format.



The screenshot displays the HuBase database interface. The top navigation bar includes "PDX model", "Gene", "Search", "Help", and "Contact Us". The left sidebar shows a "Gene Search" section with "KRAS" selected, and a "Tumor Models(15)" section with various cancer types listed. The main content area is titled "Data Analysis" and shows a "Mutation" table for the KRAS gene. The table lists 10 mutations, including their type, gene, PDX model, model type, mRNA_pos, dbSNP_id, CDS_pos, Ref, Alt, Depth(Ref/Alt), Amino_acid_change, and Confidence. An "Export" button is visible above the table.

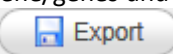
Mutation type	Gene	PDX model	Model type	mRNA_pos	dbSNP_id	CDS_pos	Ref	Alt	Depth(Ref/Alt)	Amino_acid_change	Confidence
nonsynonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	99	rs121913529	35	G	A	8/18	G13D	PASS
nonsynonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	115	-	451	A	G	22/21	R151G	SnpCluster
nonsynonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	117	-	453	A	T	20/6	R151G	SnpCluster
synonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	520	-	456	G	T	18/6	V152V	SnpCluster
nonsynonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	523	-	459	G	T	18/6	E153D	SnpCluster
synonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	529	-	465	T	C	19/6	A155A	SnpCluster
synonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	532	-	468	T	C	19/6	F156F	SnpCluster
synonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	541	-	477	G	A	21/5	L159L	LowQD
synonymous mutation	KRAS	CR0892-P7	Colorectal Cancer	547	rs4362222	483	G	A	122	R161R	PASS
InUTRregion	KRAS	CR0892-P7	Colorectal Cancer	707	rs1137282	-	T	C	2/26	-	PASS

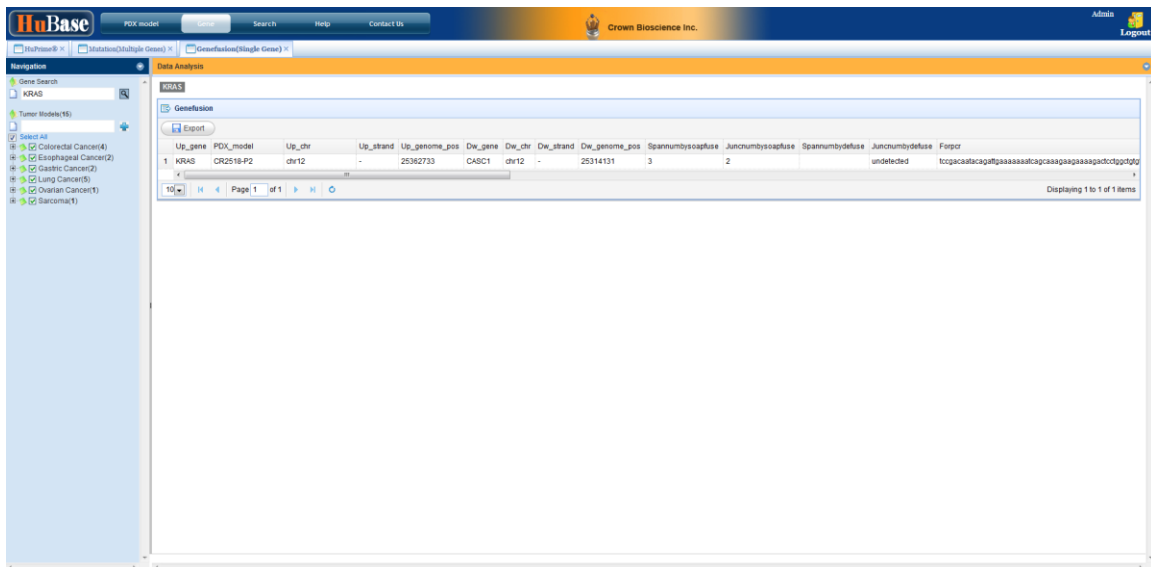
You can select mutation type by selecting on the column header of "Mutation Type" as your research need.

By clicking one specific dbSNP_id, you can jump to links and get more detailed information.

	Mutation type	Gene	dbSNP_id
	▼		
1	Select All	KRAS	
2	inUTRregion	KRAS	rs712
3	nonsynonymousmutation	KRAS	rs4285970
4	synonymousmutation	KRAS	rs1137282
5		KRAS	rs4285970
6		KRAS	rs712
7		KRAS	rs4285970
8		KRAS	-
9	synonymousmutation	KRAS	rs712
10	inUTRregion	KRAS	rs4285970
			rs712

9. Gene-Genefusion

Click "Gene->Genefusion->Single Gene" or "Gene->Genefusion->Multi Genes" from the navigation bar, select a gene/genes and PDX models, click  to display gene related fusion information, and you can click  to export result in xls format.

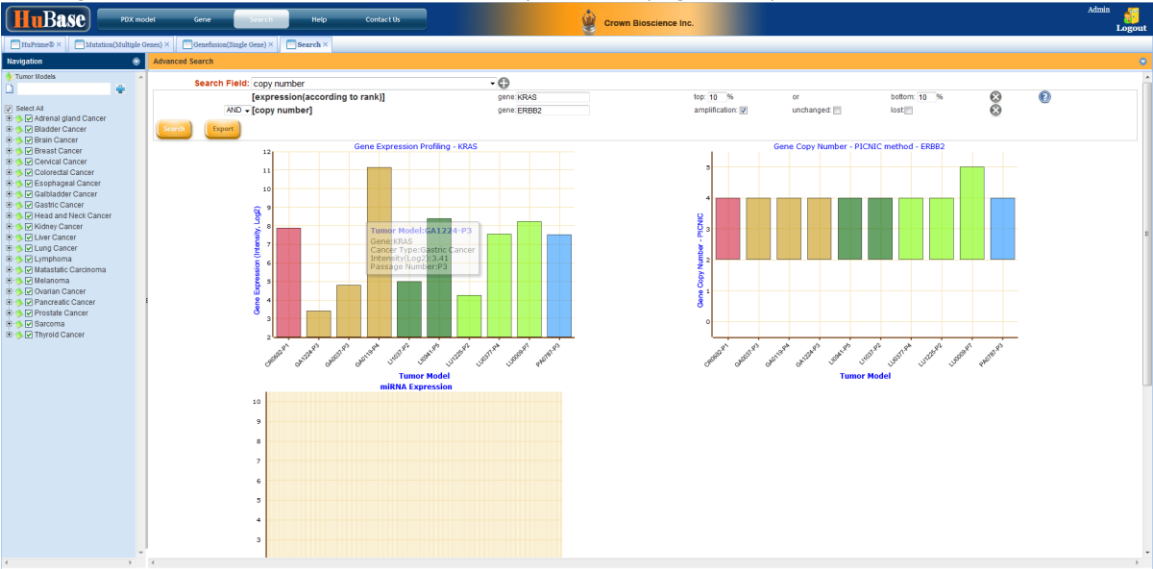


The screenshot shows the InBase web application interface. The navigation bar includes "Gene Search", "PDX model", "Search", "Help", and "Contact Us". The "Gene Search" section is active, showing a list of genes and tumor models. The "Gene-Genefusion" section is selected, displaying a table of gene fusion data for KRAS. The table has columns for Up_gene, PDX_model, Up_chr, Up_strand, Up_genome_pos, Dn_gene, Dn_chr, Dn_strand, Dn_genome_pos, Spannumbycapfuse, Juncnumbycapfuse, Spannumbydefuse, Juncnumbydefuse, and Fgpr. The table displays one row of data for KRAS in PDX model CR2516-P2, located on chromosome 12 at position 25362733. The table also shows the sequence of the fusion: tcgcacatcagatgaaataatcagaaagagaaagatctctgggg. The interface includes a search bar, a navigation bar, and a table of gene fusion data for KRAS. The table has columns for Up_gene, PDX_model, Up_chr, Up_strand, Up_genome_pos, Dn_gene, Dn_chr, Dn_strand, Dn_genome_pos, Spannumbycapfuse, Juncnumbycapfuse, Spannumbydefuse, Juncnumbydefuse, and Fgpr. The table displays one row of data for KRAS in PDX model CR2516-P2, located on chromosome 12 at position 25362733. The table also shows the sequence of the fusion: tcgcacatcagatgaaataatcagaaagagaaagatctctgggg.

Finally, the search options can be combined using the "AND" or "OR" menu. If "AND" menu was selected, then only the model that satisfies all the search options will be selected. If "OR" menu was selected, any model that satisfies any one search option will be selected.

After clicking the "Search" Button, the models that meet the options will be output on the page, and

they can be downloaded in xls format by clicking the button of . Clicking the model in the result table will lead you to the page of Huprime.



PDX model	KRAS Expression	ERBB2 CopyNumber
CR0692-P1	7.883966	4
GA0037-P3	4.791383	4
GA0119-P4	11.155722	4
GA1224-P3	3.413406	4
LU0941-P5	8.379888	4
LI1037-P2	4.996033	4
LU0009-P7	8.242235	5
LU0377-P4	7.558628	4
LU1225-P2	4.255745	4
PA0787-P3	7.518081	4

20 About 10 results

11. Help-About HuBase

Click "About HuBase" from the navigation bar, the page briefly describes the main content of this website.

Version: 2.0

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note:

This website displays pharmacological and genomic information of in house PDX models in Crownbio Bioscience Inc. The genomic resource includes the gene expression profiling, gene copy number, miRNA expression profiling, gene mutation and gene fusion information. The website also provides the PDX model selection function according to the users' search conditions.

12. FAQ

FAQ

1. What's the correlations between RNAseq and Microarray based gene expression level ?

The RNAseq and Microarray (Here Affymetrix U219) based gene expression are well correlated($R=0.62\sim0.75$), which is accordance with reports by other studies. you can check "Correlation_between_RNAseq_and_U219_array_expression.pdf" for detailed correlation information of PDX models.

Download: [Correlation_between_RNAseq_and_U219_array_expression.pdf](#)

[RNA-seq An assessment of technical reproducibility and comparison with gene expression arrays.pdf](#)

2. Why does not display the chart ?

13. Contact Us- Contact Us

Contact Crown Biosciences

Thank you for your interest in HuBase. We welcome your questions and comments. Please reach out to the appropriate contact below.

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Association for Assessment and Accreditation
of Laboratory Animal Care International

Questions/Inquiries?

To help us direct your questions, please select from the following categories:

☐ service/scientific questions ☐ website feedback

Type your detailed questions here

Send

14. Log out



Click the  at the top right corner to log out.